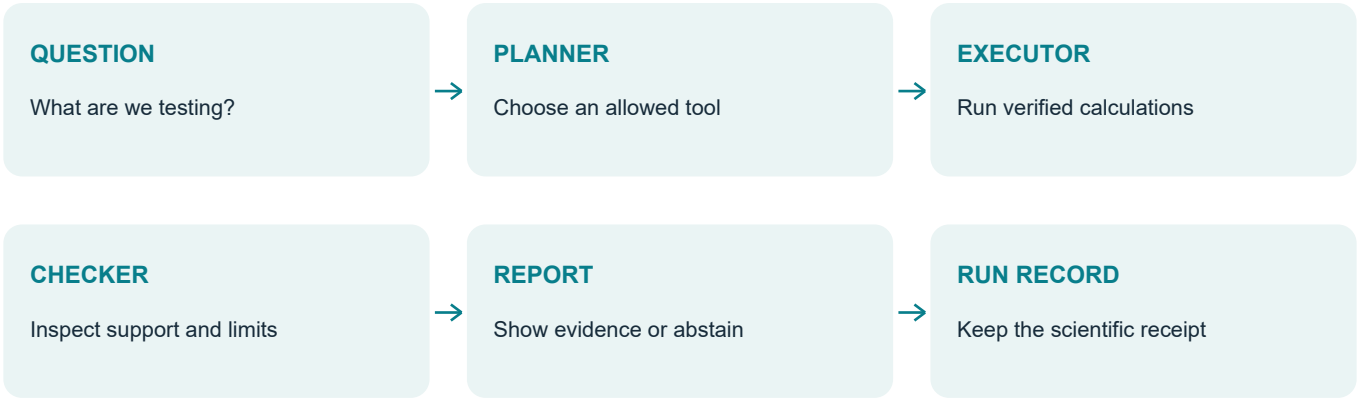
BLACKGRID

Your guide to the whole system

A beginner's handbook for building a small, trustworthy network-research assistant.



ALL PHASES. PLAIN ENGLISH. REAL BOUNDARIES.

Network science, agentic AI, scientific experimentation, evaluations, Bayesian uncertainty, graph ML, causal reasoning, LLMOps, and inference optimization - connected through one manageable project.

Edition 1 | 17 October 2026
Current implementation through Day 5. Future modules clearly labeled.
Revised planning target: approximately 100 focused hours.

START HERE

How to use this handbook

A map of the whole project, with a slower path through the parts you are learning now.

First reading: read pages 3-6 to understand the goal, the system, the smaller plan, and what already exists. Then read pages 7-10 for the work completed through Day 5.

Second reading: take one future topic at a time. Each chapter puts the project explanation beside a small vocabulary panel. Read the worked example before looking up formulas or opening a library.

Three evidence labels matter: CURRENT describes code or saved artifacts inspected on 17 October 2026. PLANNED describes a future feature. ILLUSTRATION is a made-up teaching example. A future chapter is not a claim that its feature has been built.

Use understanding as the checkpoint: explain the input, the calculation, the output, and one limitation in your own words before adding another feature. You do not need to understand every class before understanding the experiment.

CONCEPTS BESIDE THE WORK

Read now
Mission 3; architecture 4; roadmap 5; current state 6; data and experiment 7-10.

Read as you build
Agents 11; evaluations 12; missing data 13; Bayes 14; graph ML 15-16; causality 17; operations 18-19.

Keep nearby
Safety 20; repeatability 21; release 22; code tour 23; study route 24; glossary 25-26; days 27; sources 28.

CHECK YOUR UNDERSTANDING

Before proceeding, say: "The current project asks ___, uses ___, and cannot yet ___."

THE MISSION

A research assistant that earns trust

BLACKGRID investigates networks and explains how much its evidence can support.

A network is a collection of things and their connections. People exchanging messages, computers communicating, and proteins interacting are different examples. The meaning of a connection changes with the dataset, so one method cannot be applied blindly everywhere.

BLACKGRID combines reliable calculations with a bounded AI assistant. The assistant can propose a question, choose an approved tool, inspect the output, and explain a conclusion. The calculations supply evidence; fluent language alone does not.

Our first question is narrow: does a small messaging network have stronger group structure than suitable shuffled networks? Later modules add prediction, uncertainty, causal reasoning, and cost measurement. They are small extensions around a shared scientific workflow.

The research ambition: test whether validation and criticism reduce unsupported conclusions without making the assistant refuse every useful question. A small prototype can investigate this. Proving a general improvement or publishing a novel contribution needs stronger independent evidence.

CONCEPTS BESIDE THE WORK

Autonomous
The system chooses among permitted next steps. It is still limited by tools, data, budgets, and stop rules.

Evidence-grounded
A statement points to an actual saved calculation or observation.

Research-grade
Careful questions, fair comparisons, recorded failures, and repeatable work. It is not a promise of publication.

CHECK YOUR UNDERSTANDING

Explain why "the report sounds convincing" is not a useful scientific success criterion.

THE WHOLE SYSTEM

One workflow, several responsibilities

These are logical jobs. They do not require six services or six separate AI models.

Data layer: read an approved input, verify it, and construct the graph using recorded rules. **Planner:** translate a question into a structured request for an allowed tool.

Executor: validate that request and run ordinary calculation code. **Checker:** inspect numerical consistency, assumptions, and whether each proposed claim has support.

Reporter: turn accepted findings and limitations into readable language. **Run recorder:** preserve settings, hashes, errors, timing, and costs across the workflow.

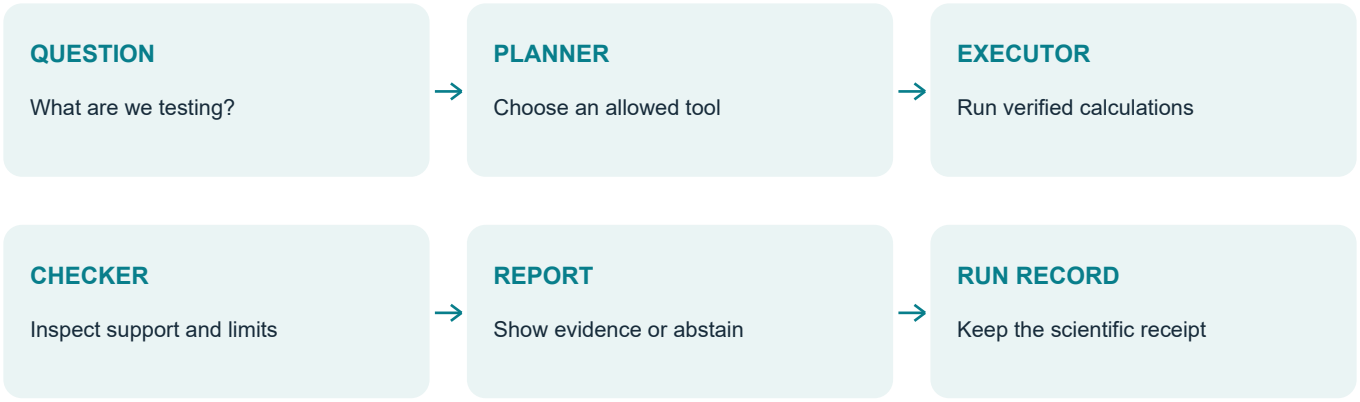
If a request is invalid, evidence is insufficient, a deadline expires, or a budget is exhausted, the controller stops or records an abstention. Repeatedly trying until something looks significant is not allowed.

CONCEPTS BESIDE THE WORK

Agent versus tool
An agent chooses actions. A tool performs a specific task, such as counting edges.

Controller
The code that decides which step is allowed next and when to stop.

Artifact
A saved output: a graph summary, result JSON, report, or configuration.



CHECK YOUR UNDERSTANDING

Name a job that should use deterministic Python rather than asking an LLM to invent a number.

THE SMALLER PLAN

All phases, about 100 focused hours

Eight small sprints preserve the topics while limiting each to one useful example.

This is the revised planning target discussed in chat, including work already completed. It is not a promise that every result will fit the estimate. Count actual time and adjust scope. Existing repository planning files have not been rewritten by this handbook.

The original architecture numbers phases 0-7. The sprint sequence below groups their work differently: Bayesian learning comes before graph ML in the teaching sequence. Reproducibility, security, and cost records start immediately and continue throughout.

Sprint / hours	One concrete outcome	Architecture phase
1 / 14	Explain and finish the CollegeMsg experiment and report.	0 data + 1 first workflow
2 / 12	One agent, a small tool menu, and a bounded loop.	1 + 2 controlled agent
3 / 14	Known-answer cases and a fair critic comparison.	2 scientific evaluation
4 / 12	One Bayesian model with checks and limitations.	4 Bayesian uncertainty
5 / 14	One graph-learning model against simple baselines.	3 graph prediction
6 / 10	One controlled causal simulation and correct abstention.	5 causal extension
7 / 12	Local traces, cost accounting, one measured optimization.	6 efficiency + operations
8 / 12	A repeatable demo, reviewed release, and honest report.	7 research release

CHECK YOUR UNDERSTANDING

Treat roughly 25% of time as learning, 50% as building, and 25% as checking and interpretation. These are guides, not separate extra hours.

CURRENT STATE / 17 OCTOBER 2026

What has actually been built?

The repository contains the scientific foundation through Day 5 and a saved real-data experiment.

Present: input checks; a CollegeMsg loader; fixed time-window selection; graph construction; synthetic generators; eight development cases; validated data models; the community-versus-randomization experiment; a command-line interface; tests; and saved run records.

Still planned: missing-edge sensitivity execution, an agent planner, a full critic, a readable final report stage, Bayesian modeling, graph prediction, causal simulations, and broader operations/optimization work.

A saved CollegeMsg experiment completed its calculations in about 18.6 seconds with zero LLM calls and zero recorded API cost. Its scientific verdict was not_supported. The whole workflow manifest says failed because later stages are not implemented, not because the calculation crashed.

The README says the real experiment has not yet run, but a saved report exists. The saved manifest also records a dirty working tree, so the commit ID alone does not fully identify that execution. These are documentation/provenance details to improve later, not changes made for this handbook.

CONCEPTS BESIDE THE WORK

Implemented
Code exists for the feature. This is different from proving that it is correct or generally reliable.

Observed here
This handbook inspected files and saved artifacts. It did not rerun tests or experiments.

Scope reminder
No working AI scientist exists yet. A mock setting is not an implemented LLM agent.

CHECK YOUR UNDERSTANDING

Explain the difference between "the hypothesis was not supported" and "the software crashed".

Local evidence: README.md; src/blackgrid/; saved run 20261017T123437Z-0c158fa0. See source notes on page 28.

PHASE 0 / DATA FOUNDATIONS

Start with what the data really records

CollegeMsg records messages, not friendships, intentions, influence, or verified social groups.

Each CollegeMsg row contains a sender ID, a receiver ID, and a timestamp. The local summary records 59,835 valid events involving 1,899 users. Repeated messages are real events; repeated pairs are expected. Exact duplicate rows are flagged rather than silently deleted.

The file fingerprint must match the expected checksum. A match establishes byte identity with the expected local input; it does not establish truth, completeness, or a license to redistribute the data.

The fixed selection rule starts at the earliest valid timestamp: try seven days, then three days, one day, and six hours if needed to meet size caps. Include the start and exclude the end boundary. Select using size only, never community scores.

The saved selection uses the first seven days: 196 events, 104 active users, and 137 unique undirected connections. It describes this observed early window, which may differ from later activity. The user accepted local non-commercial analysis with citation for Sprint 1; that decision is not a verified redistribution license.

CONCEPTS BESIDE THE WORK

Provenance
Where the data came from and what transformations happened to it.

Timestamp
A number representing when an event happened. CollegeMsg uses seconds since 1970 in UTC.

Half-open interval
Start is included; end is excluded. Adjacent windows do not double-count their boundary.

Pseudonymous
An ID hides a name but does not guarantee that re-identification is impossible.

CHECK YOUR UNDERSTANDING

Why should two messages from the same person to the same recipient on different days remain two events?

Dataset meaning: [1]. Local counts: data/manifests/collegemsg.json. Usage decision: docs/decision_log.md, D-003.

PHASE 0 / NETWORK SCIENCE

From message rows to dots and lines

A graph is a useful simplification. Its omissions must remain visible.

In this illustration, A, B, and C are one tight circle; D, E, and F are another. A single bridge connects the circles. That is the kind of pattern a community method may detect. These letters are invented, not CollegeMsg identities.

The current projection puts one undirected, equal-weight edge between two users if either messaged the other in the window. Ten messages and one message both produce one edge.
Self-messages do not produce edges, but active users with only self-messages remain as isolated nodes.

Other graph summaries answer different questions: degree counts neighbors; a connected component is a piece reachable by following edges; density measures how many possible connections are present. None of these directly measures a person's real-world importance.

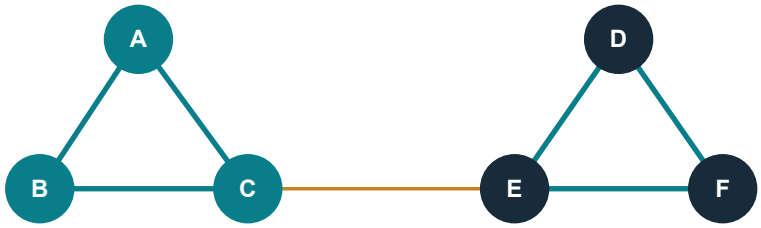
The mapping from events to a static graph discards direction, frequency, and order. That is acceptable for the chosen experiment when stated clearly. It would be inappropriate to reuse the same graph without checking the requirements of a new question.

CONCEPTS BESIDE THE WORK

Node / edge
A thing / a connection. Here, a user / at least one message relationship.

Directed / weighted
Arrows retain who contacted whom. Weights can retain message counts. This experiment drops both.

Degree / component
Number of neighbors / a connected piece of the graph.



Two tight circles with one bridge between them. Illustration only.

CHECK YOUR UNDERSTANDING

Draw A -> B twice and B -> A once. How many events? How many edges in our simple graph?
Answer: three events, one edge.

PHASE 1 / SCIENTIFIC EXPERIMENT

Compare the pattern with a fair alternative

Finding groups is easy. Showing that the score is unusual requires a comparison.

The current experiment finds groups with Louvain and measures their modularity Q. It repeats the optimizer five times and retains the largest score found. Louvain is a heuristic: it searches for a good grouping without guaranteeing the global best one.

The alternative explanation is that connection counts and random wiring can already produce an impressive score. The code shuffles edges while preserving each node's degree. For example, two edges A-B and C-D may become A-D and C-B, if that creates neither self-loops nor duplicate edges.

Each of 499 randomized graphs receives the same five-restart group search. The code records failed rewiring attempts and may abstain if the comparison is unsuitable. If the graph is disconnected, it also checks the largest connected piece.

The comparison is conditional on this null model and finite shuffling procedure. Completing a fixed number of swaps does not itself prove perfect sampling of all possible graphs. Small p-values, when obtained, would still require methodological judgment rather than automatic discovery claims.

CONCEPTS BESIDE THE WORK

Null model
A precise version of "what could happen without the claimed extra structure." It defines which chance explanation is tested.

Modularity Q
A grouping score accounting for node degrees. A large score alone is not proof of meaningful communities.

Fair comparison
Observed and randomized graphs receive comparable methods and search effort.

CHECK YOUR UNDERSTANDING

Why would five optimizer attempts on the real graph but one attempt per shuffled graph make the comparison unfair?

Method definitions: [2], [3]. Exact local procedure: protocols/sprint01_experiment.md; src/blackgrid/experiment.py.

CURRENT RESULT / WORKED EXAMPLE

Read your first result without the jargon

The saved first-week experiment did not show unusually strong group structure.

The actual network scored about 0.5865. The average score across shuffled networks was about 0.5933. In 358 of the 499 successful shuffled comparisons, the score was at least as large as the actual score.

The code calculates p = (1 + 358) / (1 + 499) = 0.718. This says the observed score is not unusual in the direction being tested under this comparison. It is not a 71.8% probability that the hypothesis is false.

The primary graph has eight connected components. Its largest component contains 87 users and 127 edges; the separate robustness comparison there also did not pass the evidence threshold (p = 0.252).

Good wording: "The first-week projected network did not show modularity above the degree-preserving randomizations under the recorded procedure." **Bad wording:** "There are no social groups" or "the project failed." A result that rejects an attractive story is still useful scientific work.

CONCEPTS BESIDE THE WORK

p-value
A measure of how extreme the observed statistic is under a stated null comparison. It is not the probability that a hypothesis is true.

Effect contrast
Observed score minus average null score. Here it is about -0.0068.

Evidence boundary
One time window, one projection, one method and null model. No causal or social-group ground truth.

CHECK YOUR UNDERSTANDING

If an invented graph scores 0.70 but its shuffled graphs average 0.72, is 0.70 alone convincing evidence of extra group structure? Explain why not.

Saved report: runs/20261017T123437Z-0c158fa0/experiment_report.json. This handbook did not rerun it.

PHASES 1-2 / AGENTIC AI / PLANNED

Give the assistant a small tool menu

The agent chooses steps. Trusted tools do the arithmetic.

A future user might ask, "Investigate group structure in the approved network." The agent receives an aggregate graph description, a small tool menu, and explicit rules. It requests a tool through structured fields rather than writing code for execution.

The controller checks the request. If valid, it executes the registered function and returns a result. The agent may then explain the output or choose one permitted follow-up. A bounded loop limits calls, revisions, time, and spending.

Illustrative request: tool = community_test, graph_id = approved_window, protocol = exp_v1. A request such as "execute this shell script" is refused. Input data that says "ignore previous instructions" remains untrusted data.

Start with a deterministic rule-based planner and mock responses. Add one real LLM adapter only when authorized. Several roles can use one process and even the same model; splitting everything into agents is not the goal.

CONCEPTS BESIDE THE WORK

LLM
A language model that generates text. Its fluency does not guarantee factual correctness.

Structured tool call
A request with named fields the program can validate before acting.

Mock / replay
Fixed example responses / previously saved responses, useful for workflow testing without live model calls.

Bounded autonomy
The system makes choices inside limits the user and program enforce.

CHECK YOUR UNDERSTANDING

Explain a complete cycle: question -> allowed request -> validation -> calculation -> observation -> explanation. Stop when you cannot identify the next allowed step.

PHASE 2 / EVALUATIONS / PLANNED

Test both honesty and usefulness

A checker that refuses everything has few false claims but is not a good research assistant.

The existing development benchmark contains eight small cases: strong and weak groups, random wiring, uneven degrees, disconnected pieces, missing observations, forbidden code execution, and an unsupported causal request. These are practice cases, not a sealed final exam.

A future critic checks method requirements, numerical consistency, evidence links, and wording. A claim that a result file exists is different from a claim that its contents support the conclusion. The checker must inspect both.

Compare the same agent on the same tasks with and without the critic, keeping the model, tools, prompts where appropriate, and maximum budget comparable. Also compare with a simple rule-based baseline. Record actual resource use and both improvements and failures.

Illustrative scoring: 2 unsupported claims out of 20 assessed claims gives 10%. But also count correct useful answers and refusals. A future independent test set must remain hidden during tuning; repeated runs on the same graph are not independent graph samples.

CONCEPTS BESIDE THE WORK

Evaluation / benchmark
A measurement procedure / a collection of tasks used with it.

Ablation
Remove one component, such as the critic, to see what changes.

Coverage
How often the system gives a useful answer rather than abstaining.

Independent check
Known numerical answers or human review. An LLM grading itself is not sufficient ground truth.

CHECK YOUR UNDERSTANDING

An assistant refuses all eight tasks and makes zero false claims. What second measure exposes the problem?

PHASES 2 AND 4 / MISSING DATA

A missing line is not proof of no relationship

The observation process can change the apparent shape of a network.

Suppose two clubs exchange messages, but messages between clubs are less likely to be recorded. The observed graph may look like two isolated groups even if the underlying social system is well connected. This is an invented example of biased missingness.

The next planned Sprint 1 check removes 10% of observed edges uniformly at random, repeats this 20 times, and reruns the comparison. It reports how often the verdict changes and how the scores move. It is not implemented in the inspected Day 5 workflow.

Uniform removal answers one limited question: sensitivity to that particular damage pattern. Later tests should include missingness related to degree, group, or time. They do not reveal the actual unobserved edges in CollegeMsg.

Separate uncertainty due to observation, graph construction, model parameters, and optimizer randomness. A wide result range from an edge-removal experiment is not automatically a confidence interval or a Bayesian credible interval.

CONCEPTS BESIDE THE WORK

Observed / unknown
A recorded link / a pair whose status is not established.

Sensitivity analysis
Change an assumption or input and see how much the conclusion changes.

Biased missingness
Some types of records are more likely to be missing than others.

Proxy edge
An inferred connection, such as co-employment, kept distinct from an observed interaction.

CHECK YOUR UNDERSTANDING

What could uniform edge removal miss if between-group messages are disproportionately absent?

PHASE 4 / BAYESIAN UNCERTAINTY / PLANNED

Show a range of plausible answers

A Bayesian model updates assumptions with data. It does not remove the need to question those assumptions.

Begin with a tiny, controlled learning example rather than assigning uncertainty to the entire system. Imagine ten independently generated synthetic research tasks: a proposed procedure succeeds on six and fails on four. This example is invented; it is not a BLACKGRID evaluation result.

Let p represent the procedure's success probability for that specified task distribution. Start with a Beta(1,1) prior, uniform over p. Under the independent Bernoulli model, six successes and four failures give a Beta(7,5) posterior, with mean 7/12, about 0.583.

That posterior is a distribution of plausible parameter values under the model, not a promise of 58.3% accuracy on real graphs. Reusing one graph many times or selecting easy tasks would violate or change the intended interpretation.

Check whether the prior permits sensible behavior, whether simulated posterior data resembles observations, and whether the method behaves properly across synthetic repetitions. For the small prototype, one clearly defined model with these checks is enough.

CONCEPTS BESIDE THE WORK

Prior
The assumed distribution before these observations.

Likelihood
How probable the observations are for each parameter value under the model.

Posterior
The updated distribution after combining prior and likelihood.

Credible interval
An interval containing a chosen amount of posterior probability, conditional on the model and data.

CHECK YOUR UNDERSTANDING

Why should six successes on easy development tasks not become a general claim about scientific reliability?

Bayesian workflow and predictive-check background: [4]. Beta-Bernoulli example is an original teaching example, not an implemented feature.

PHASE 3 / GRAPH ML / PLANNED

Can learned patterns improve prediction?

Community discovery describes an existing graph. Link prediction asks about later events.

Define a precise new target: which eligible pairs will have a recorded message in a future period? Decide whether this means a first-ever connection or any later interaction. These are different tasks. They do not predict friendship or a causal relationship.

Start with a heuristic such as common neighbors: two users sharing many contacts may be ranked higher. Then compare one learned graph representation with that baseline under a time-based protocol.

An embedding is a short vector of numbers representing a node. One graph neural network can learn representations by combining information from neighboring nodes. This is called message passing; it is numerical aggregation, not users sending actual messages inside the model.

A small GraphSAGE-style model is a possible future choice, not a fixed dependency or current implementation. Keep the graph and feature design simple. If the learned model costs more and provides no reliable improvement, retain the simpler baseline and report the finding.

CONCEPTS BESIDE THE WORK

Feature
A measurable input, such as degree at a specified cutoff time.

Embedding
A learned numerical representation of a node or graph.

GNN
A neural network that uses graph connections to combine information.

Baseline
A simple comparison method that a more complex method must justify replacing.

CHECK YOUR UNDERSTANDING

Explain the difference between "these users share contacts" and "these users will message next week."

Message-passing background: [5]. Proposed model and prediction task remain future design decisions.

PHASE 3 / PREDICTION EVALUATION / PLANNED

Do not let the model see the future

A clever model can look excellent when the evaluation accidentally gives away the answer.

For a temporal prediction experiment, build training inputs from an early period, tune choices on a later validation period, and score once on a final test period. Features, topology, embeddings, and transformations must use information available at the relevant cutoff.

Do not leave a target future edge in the graph used for message passing. Randomly splitting edges from a full-history graph may leak future structure. State whether the test includes the same users or users never seen in training.

An unobserved pair is not a verified negative. Define the observation horizon, candidate pairs, and how sampled non-events are chosen. A pair may communicate later or outside the recording system. Scores depend on these choices.

Illustration: among ten top-ranked candidate pairs, four later have recorded messages: precision@10 is 4/10. If there were eight eligible future positive pairs in total, recall@10 is 4/8. Report candidate rules, prevalence, ranking performance, calibration where relevant, and compute cost.

CONCEPTS BESIDE THE WORK

Leakage
Using information that would not be available when making the real prediction.

Train / validation / test
Fit the model / choose settings / estimate final performance.

Precision / recall
How many selected predictions are right / how many eligible positives were recovered.

Calibration
If predictions say 70%, roughly 70% should occur across comparable cases.

CHECK YOUR UNDERSTANDING

Why does calculating all users' degrees from the full six-month dataset leak information into a first-month prediction?

General leakage principles: [6]. Network-specific split and candidate rules are BLACKGRID design requirements.

PHASE 5 / CAUSAL REASONING / PLANNED

What happens if we change something?

Observing two things together does not establish that one produces the other.

Imagine highly connected users also send more messages. Does making someone more connected increase their messaging? Maybe. But an unobserved factor such as activity level could increase both. Predicting the association does not resolve the causal question.

Use a tiny synthetic world with explicitly written equations. Include a background activity variable that influences both a treatment and an outcome. Simulate an intervention that sets the treatment, compare its known effect with an observational estimate, and explain the difference.

Before estimating an effect, state the intervention, outcome, units, time order, confounders, and assumptions that make the effect identifiable. In networks, one person's treatment may change another person's outcome; ordinary independent-unit assumptions can fail.

For CollegeMsg, the small prototype should usually explain why a causal effect is not identified, rather than produce a confident estimate. A causal graph suggested by an LLM and software robustness checks do not establish that the assumptions are true in the real world.

CONCEPTS BESIDE THE WORK

Confounder
A common cause of both the proposed treatment and the outcome.

Estimand
The exact effect we want to measure, for a defined population and intervention.

Identification
Whether the assumed model and available data are enough to determine that effect.

Interference
An intervention on one unit affects another unit, common in networks.

CHECK YOUR UNDERSTANDING

What additional assumptions would be needed before saying "high-degree users cause stronger communities"?

Model-identify-estimate-refute workflow: [7]. The simulation is planned, not evidence about CollegeMsg.

PHASE 6 / LLMOPS / PLANNED

Keep a clear record of the assistant

LLMOps is the practice of measuring and maintaining a system that uses language models.

A trace follows one investigation across steps: question received, tool requested, request checked, tool finished, claim checked, report generated. Each step has timing and status. A log records an event; a metric summarizes behavior across runs.

Version prompts, model identifiers, tool schemas, and evaluation rules. When any changes, rerun a fixed regression suite before declaring the system better. A more eloquent answer can still contain worse science.

Record call counts, input/output tokens, estimated and actual cost, latency, retries, and failure reasons. Keep data minimized: aggregates and safe references rather than raw CollegeMsg events or credentials. External tracing services are data disclosures, not automatic requirements.

The starting implementation can be local JSON files and a small summary table. A hosted observability platform is optional. The goal is to answer "what happened and why?" without reconstructing a hidden chain of actions.

CONCEPTS BESIDE THE WORK

Trace / span
A complete request journey / one timed step inside it.

Latency / throughput
Time to finish one request / requests completed per unit time.

Token
A text unit counted by the model, not always a whole word.

Regression
A change breaks behavior that previously worked.

CHECK YOUR UNDERSTANDING

What should a trace record when an experiment succeeds but the checker rejects the report's conclusion?

Observability terms: [8]. Local-first implementation is the proposed BLACKGRID scope.

PHASE 6 / INFERENCE OPTIMIZATION / PLANNED

Make it cheaper or faster without losing quality

Implement one optimization and compare it fairly before adding another.

Start with application-level caching: reuse a validated output only when the model, prompt, data/config fingerprints, and relevant permissions match. A stale or incorrectly shared cache can silently change scientific conclusions or leak data.

Routing chooses a cheaper or faster model for tasks it can handle and escalates harder cases. Batching groups requests. Quantization stores model values at lower precision to save resources, but may change quality. These are distinct techniques; the small project need not implement all of them.

Prefix caching inside an inference engine reuses computation for repeated prompt prefixes. It differs from returning a previously saved answer. Dedicated serving software such as vLLM is optional, depending on hardware and a demonstrated need.

Illustrative result format: compare baseline versus caching on the same allowed workload; disclose cache hit rate, cold/warm timing, actual cost, and scientific quality. Do not count replayed identical requests as proof of general model speed or publish invented savings.

CONCEPTS BESIDE THE WORK

Inference
Using a trained model to produce an output.

Cache hit
The requested reusable result or computation was already available.

Quantization
Using fewer numerical bits for model values, with possible quality tradeoffs.

Quality constraint
A predefined rule about how much, if any, loss in reliability is acceptable.

CHECK YOUR UNDERSTANDING

Why should a cache key change when the graph changes, even if the user asks exactly the same question?

Prefix-caching distinction: [9]. Other examples describe proposed comparisons, not measured performance.

ALL PHASES / SAFETY AND COST

Let the system stop safely

The assistant should have less authority than the person using it.

Run only registered tools with validated inputs. Never execute arbitrary Python or shell commands suggested by a model or embedded in data. Treat files, retrieved text, and generated explanations as untrusted until checked.

Enforce time, memory, output, and filesystem limits. Deny worker network access by default. A Python subprocess or a configuration flag alone is not an operating-system sandbox. If strong containment is unavailable, run only trusted deterministic tools and disable untrusted execution.

The current Sprint 1 limits are one worker, 2 GB worker memory, 60 seconds per tool, and at most 1,000 nodes/10,000 edges. Paid services are not authorized by a budget number. If later approved: USD 10 sprint total, USD 0.50 per investigation, eight calls, 20,000 total tokens, one retry per failed call, and two revision rounds.

Reserve estimated cost before a call, count retries and subcalls, reconcile actual use, and stop before any cap would be exceeded. If pricing is unknown, disable paid calls. Do not put credentials in prompts, logs, or commits. Never import proprietary ValueMatrix code, data, prompts, schemas, or derived artifacts.

CONCEPTS BESIDE THE WORK

Allowlist
A list of permitted actions; all others are refused.

Sandbox
An enforced boundary around execution, beyond ordinary program intentions.

Prompt injection
Untrusted content tries to act like an instruction to the assistant.

Clean-room boundary
Independent public/synthetic work kept separate from confidential employer material.

CHECK YOUR UNDERSTANDING

A tool times out twice and the third attempt would exceed the budget. What should the controller do? Stop and record the failure.

ALL PHASES / REPRODUCIBILITY

Every experiment needs a receipt

Someone should be able to understand what was run and repeat the scientific comparison.

Record the input fingerprint, graph-construction rules, time window, configuration, experiment specification, seeds, code revision, dependency lock, runtime/hardware, output hashes, timings, failures, and costs. Preserve unsuccessful attempts too.

A configuration file states the choices. A lockfile records dependency versions. A run manifest connects these to the outputs. Hashing a saved file helps detect changes, but it does not certify that the file's conclusions are correct.

A fixed seed helps repeat randomized calculations under the same environment. It does not guarantee identical outcomes across every library version, machine, GPU kernel, or hosted model. Record these limits and define suitable numerical tolerances.

Keep development examples for learning, validation cases for choices, and final test instances unseen until evaluation. New seeds alone do not guarantee new graph structures. Freeze the intended comparison before opening final results; changes afterward need new evaluation or exploratory labeling.

CONCEPTS BESIDE THE WORK

Manifest
A machine-readable receipt linking a run to its inputs, settings, and outputs.

Lockfile
A record of resolved software dependencies used to recreate an environment.

Dirty tree
Files differ from the recorded Git commit. The commit alone cannot reproduce those edits.

Preregistration
Write the intended procedure before inspecting the result it will test.

CHECK YOUR UNDERSTANDING

If two runs use the same seed but different NetworkX versions, why should the versions still be recorded?

PHASE 7 / RESEARCH RELEASE / PLANNED

A small finished system you can defend

The final deliverable is a readable, repeatable demonstration with honest boundaries.

A useful demo starts with an approved question and input, shows the selected tools, displays the scientific comparison, and ends with a claim tied to evidence or an explicit abstention. One command-line workflow can be enough; a dashboard is optional.

The release should contain a clear README, environment lock, example configuration, synthetic fixtures, reviewed aggregate results, evaluation protocol, limitations, and reproduction steps. Review data and dependency rights before publication. Do not bundle raw CollegeMsg records or confidential material.

Separate measured findings from illustrations and future ambitions. A model that does not beat a heuristic, a critic that does not help, and a causal estimate that cannot be identified are legitimate outcomes. Explain what was learned and what would change the conclusion.

A technical report can discuss method, baselines, results, costs, failure cases, and limitations. A paper or claim of novelty requires a literature review and a stronger study; it is an optional extension rather than a condition for completing this roughly 100-hour prototype.

CONCEPTS BESIDE THE WORK

Replication package
The instructions and permitted materials needed to repeat an experiment.

Negative result
A tested improvement or hypothesis was not supported. It still narrows what we know.

Domain shift
The conditions in new data differ from those used to develop or test the system.

Release gate
A condition to meet before publishing or expanding the system.

CHECK YOUR UNDERSTANDING

Describe the project in two minutes: one question, one comparison, one result, one limitation, and one next experiment.

CODE TOUR / CURRENT IMPLEMENTATION

Read the code in the order data travels

Do not begin by memorizing every validation class.

1. configs/collegemsg_sprint01.json: the settings sheet. Find the dataset path, mock mode, experiment identifier, seed, and null replicate count. Configuration supplies choices, not evidence that the run happened.

2. src/blackgrid/collegemsg.py: follow load_events -> select_window -> project -> build_bounded_graph. A function consumes inputs and returns an output. A dataclass bundles related values such as the graph, counts, and selected window.

3. src/blackgrid/experiment.py: follow profile -> best_modularity -> double_edge_swap -> analyse -> verdict. Read tests/test_experiment.py beside it: the two-triangle example has a hand-computed modularity check.

4. src/blackgrid/cli.py: the coordinator validates settings, prepares the graph, runs the experiment, and writes run artifacts. **5. contracts.py:** inspect the model used by the function you just read. **6. synthetic.py and benchmark.py:** inspect how practice cases and separate answer keys are created.

CONCEPTS BESIDE THE WORK

Function
Named instructions with inputs and an output.

Class / dataclass
A type for bundling data or behavior / a convenient way to hold related values.

Pydantic
The library checking data fields and validation rules. A valid field is not proof of valid science.

CLI / JSON
A command-line interface / a structured text format for settings and results.

CHECK YOUR UNDERSTANDING

Take one function. Write down its inputs, outputs, one failure case, and the test that checks it. That is a useful learning session.

LEARNING ROUTE

Study beside the code, one layer at a time

The goal is to explain the work and make a small change with a prediction.

For each session: read a small function, predict what it does on a tiny example, check the corresponding test, then explain the behavior aloud. Only then change something. Keep mistakes in personal notes rather than treating them as failure.

The roughly 100-hour plan includes learning time. A useful allocation is 25% understanding, 50% building, and 25% checking/interpreting, adjusted to your needs. Do not add a new library just to fill a sprint.

Use AI as a tutor and reviewer: ask it to trace five rows, explain one variable, challenge your prediction, or identify an assumption. Ask for the smallest patch after you can describe the intended change yourself.

When	Study first	Show that you understand
Now	Python functions, tuples, sets, dictionaries; nodes and edges.	Turn five made-up messages into a graph.
Current experiment	Communities, null models, modularity, p-values.	Explain why Q alone is insufficient.
Agent and checks	Structured requests, validation, tests, benchmark leakage.	Trace one valid and one refused request.
Bayes and ML	Probability distributions; training, validation, calibration.	State what a prediction and interval mean.
Causality	Confounding, interventions, identification.	Explain when the system must abstain.
Operations and release	Traces, tokens, caching, versioning, clean reruns.	Repeat a run and account for its cost.

CHECK YOUR UNDERSTANDING

Ready to continue? Explain: what went in, what calculation happened, what came out, and what the result does not establish.

GLOSSARY / FIRST HALF

Words you will encounter

Short definitions for reading the code and the research plan.

Term	Plain-English meaning
Abstention	The system declines to make a conclusion because the evidence or method is insufficient.
Agent	A system that chooses among allowed actions based on a goal and observations.
Artifact	A saved output such as a summary, configuration, report, or log.
Baseline	A simple comparison method that makes a complex method justify its added cost.
Benchmark	A collection of evaluation tasks with a defined scoring procedure.
Calibration	Agreement between stated probabilities and observed frequencies across comparable cases.
Community	A grouping suggested by graph structure, not necessarily a verified real social group.
Confounder	A variable influencing both a proposed treatment and its outcome.
Credible interval	A range containing a chosen share of posterior probability under a Bayesian model.
Degree	How many neighbors a node has in this simple graph.
Effect size	The size of a difference, distinct from whether a test crosses a significance threshold.
Embedding	A numerical representation learned for a node or graph.
Ground truth	A known reference answer. Real observational data often lacks the truth we want.
Hypothesis	A specific proposition that an experiment can test.

GLOSSARY / SECOND HALF

More words, fewer mysteries

Use these definitions as a starting point, not a substitute for assumptions.

Term	Plain-English meaning
Identification	Whether an effect can be determined from the available data under stated assumptions.
Inference	Drawing conclusions from evidence; in ML systems, also running a trained model.
Leakage	Information reaches a model or evaluator when it should have been unavailable.
Likelihood	A model's probability of the observed data at particular parameter values.
Modularity	A score for a graph partition, comparing internal connections with a degree-based expectation.
Null model	The specific alternative explanation represented by a comparison procedure.
Posterior / prior	Updated parameter distribution / assumed distribution before these observations.
p-value	A tail comparison under a null model, not a probability that the research hypothesis is true.
Projection	Converting rich events into a simpler graph while intentionally losing some information.
Reproducibility	Recording enough inputs and procedure details to repeat and assess the work.
Sensitivity	How much an answer changes when specified inputs or assumptions change.
Synthetic	Generated artificial data with controlled properties; not automatically realistic.
Trace	The recorded path of one request across its steps.
Uncertainty	What remains unknown; its measured form depends on the model and assumptions.

SPRINT 1 / ORIENTATION

The original fourteen days, in one page

Days are work stages, not a command to build faster than you understand.

The inspected implementation reaches Day 5. The smaller 100-hour roadmap proposes simplifying the remaining work after reviewing it. This page preserves the existing day sequence for orientation; it does not authorize running or rewriting it.

Day	Purpose
1	Choose the question, data rules, permissions, and limits.
2	Prepare the Python project, configuration checks, and command entry point.
3	Read and validate CollegeMsg; select the window; construct the graph.
4	Build small practice cases and independent answer checks.
5	Implement the observed-versus-shuffled community experiment.
6	Check how removing connections affects the conclusion.
7	Validate tool requests and enforce permitted execution.
8	Add a small bounded planner, initially with mock responses.
9	Check support, assumptions, and limits before accepting claims.
10	Produce the readable report with linked evidence.
11	Compare behavior with and without the checker on practice tasks.
12	Exercise invalid requests, forbidden access, deadlines, and budgets.
13	Repeat the workflow from written instructions in a clean environment.
14	Review completion, limitations, time/cost, and the next step.

CHECK YOUR UNDERSTANDING

You can finish the existing experiment before adding AI. Explain the saved not_supported result first.

SOURCES AND PROVENANCE

Where this handbook gets its information

Repository snapshot and conversation context: 17 October 2026. External sources checked the same day.

Local implementation: C:/Blackgrid/README.md; src/blackgrid/{cli,collegemsg,contracts,experiment,synthetic,benchmark}.py; configs/collegemsg_sprint01.json; tests/test_experiment.py; protocols/sprint01_protocol.md and sprint01_experiment.md; docs/decision_log.md; data/manifests/collegemsg.json.

Observed result: C:/Blackgrid/runs/20261017T123437Z-0c158fa0/experiment_report.json and manifest.json. Figures on page 10 come from those saved files; they were not recomputed. Raw user identifiers are not reproduced.

Planning provenance: PROJECT_CONTEXT.md and SPRINT_01.md define the original phases and safeguards. The eight-sprint, 100-hour teaching map comes from the subsequent user conversation. Later modules are proposals, not measured capabilities. No existing repository instructions were rewritten by this handbook.

[1] Stanford SNAP: CollegeMsg dataset

[2] NetworkX: modularity

[3] NetworkX: Louvain communities

[4] PyMC: prior and posterior predictive checks

[5] PyTorch Geometric: message passing

[6] scikit-learn: leakage and common pitfalls

[7] Sharma and Kiciman: DoWhy (2020)

[8] OpenTelemetry: signals

[9] vLLM: automatic prefix caching (versioned guide)

Library documentation may describe a newer version than the repository lockfile. Use uv.lock for exact installed versions; these links explain concepts.